Final Project

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Introduction

Patient tracking and scheduling is extremely important in a healthcare system. It allows clinical and administrative staff to know where a patient is located within the facility at any given time. Keeping this information accurate is important because it directly impacts patient safety and also improves the workflow of the staff responsible for patient care. When a patient's location and status are tracked, departments are able to coordinate more efficiently and respond quickly to patient needs.

Many organizations still rely on outdated systems and in some cases even manual processes such as spreadsheets or paper documentation to track patients and manage the scheduling. But these older processes will lead to a number of issues including longer patient wait times, scheduling errors, and workflow issues between departments. When a patient's information is not updated in real time, the staff may struggle to locate patients, assign available beds, or properly coordinate procedures.

To address these issues, this IS project focuses on the implementation of a patient tracking and scheduling system. The purpose of this system is to provide real-time visibility into patient location, movement throughout the hospital, and overall bed availability. With this system in place, hospital staff will be able to quickly see where patients are located, determine which beds are available, and coordinate patient movement more efficiently across the departments.

The system will also support scheduling functionality that allows staff to assign patients to available beds and manage upcoming procedures or treatments. By improving the accuracy and visibility of patient information, the hospital will be able to reduce delays, improve

communication between teams, and support a more efficient workflow for both clinical and administrative staff.

System Development Life Cycle

The System Development Life Cycle (SDLC) is a framework used for designing and implementing information systems. By following the SDLC, an organization is able to plan projects in a more structured way, use resources more efficiently, and ensure that the patient tracking and scheduling system meets the needs of its stakeholders. Following this framework also helps reduce risks throughout the project.

Typically, the SDLC includes several phases such as planning, analysis, design, development, implementation, and maintenance. Each phase plays an important role in guiding the project team and helping the organization reach its goals throughout the implementation process. Having these phases in place also helps reduce the chance of errors and keeps the project organized.

During the first phase of the SDLC, the planning phase, the team will first identify the issue that needs to be solved. In this case, the hospital has identified that they are experiencing issues with their current patient tracking and scheduling process. They are currently using a manual and outdated system that needs to be replaced in order to improve workflow and provide better visibility into patient location and available beds throughout the hospital.

The analysis phase is where the project team gathers information and requirements for the new system that will be implemented. During this phase, the team will study the hospital's workflows and meet with stakeholders to understand what they expect from the new system. In this project, the hospital is looking for a system that can track patient location, monitor bed availability, and allow staff to schedule patients appropriately for procedures and treatments.

During the design phase, IT professionals such as system analysts, engineers, and architects will determine how the system will function and what integrations may be required for the system to work properly. This phase focuses on designing the structure of the system, including databases, interfaces, and system architecture. The goal of this phase is to ensure the system is designed properly before development begins so that there are fewer issues when the system goes live.

Once the design phase is complete, the project moves into the development phase. During development, the system is actually built based on the requirements identified during the analysis phase. Developers will create the application, build the necessary databases, and integrate other systems that may be involved, such as the hospital's electronic health record (EHR) system if needed.

After the system is developed, the project moves into the implementation phase. This phase includes testing the system's functionality, migrating any existing data into the new system, and installing the system in the live environment. Because this is a critical phase of the SDLC, the project team will provide strong support to the organization during this time. This will include training staff members who will be using the system and ensuring that the system operates correctly before full adoption.

The final phase of the SDLC is maintenance. Once the system is operational in the hospital environment, it must be monitored and maintained. Subject matter experts (SMEs) will oversee the system, address any issues that arise, and make improvements when necessary. Over time, updates and enhancements may be made based on feedback from users to ensure the system continues to meet the hospital's needs.

Development Methodology

The Waterfall methodology will be used for the implementation of this project. Overall, this approach will be the easiest for our team to follow because it is simple and structured. In the Waterfall model, each phase of the SDLC must be completed before moving on to the next phase. This works well for a project of this size because the system requirements are defined before development work begins.

By using the Waterfall methodology, the project team can ensure that stakeholder needs are fully understood before the system is actually developed. This can help prevent additional work later in the project because the team will already have a clear understanding of what the system is expected to do. Learning and documenting hospital workflows during the early phases of the project will also help guide development and ensure that the new system supports the way staff already perform their daily tasks.

This methodology also places a strong emphasis on documentation and planning. In healthcare environments, it is important that new systems are implemented with minimal disruption to staff workflow. If a system is poorly planned or deployed incorrectly, it can create confusion for staff and may even impact patient care. Using a structured methodology like Waterfall helps reduce these risks by ensuring that each phase of the project is carefully planned and completed before moving forward.

Overall, the Waterfall methodology provides a clear and organized approach for implementing the patient tracking and scheduling system. By following this structured process, the project team can ensure that the system is properly planned, developed, and tested before it is fully deployed within the hospital environment.

Planning Objectives and Strategies

Our team's goal is to meet the needs of the clinical and administrative staff by providing a system that can accurately track patients throughout the hospital in real time, manage bed availability, and allow staff to schedule patients accordingly. In order to accomplish this, we will need to establish clear objectives early in the project so that the system meets the needs of the stakeholders. Proper planning will allow our team to accomplish these goals while also improving the workflow of the hospital staff.

Improving visibility of patient location throughout the hospital is a primary objective of this project. Accomplishing this objective will allow the other objectives to be achieved more easily as well. Staff members need to be able to determine exactly where a patient is located at any given time. Having real-time access to this information will help reduce confusion among staff members and departments and allow them to coordinate patient care more efficiently.

Another objective is improving the management of beds across the hospital. When a patient arrives at the facility and requires a bed, there should not be any uncertainty about where that patient should be placed. Whether it is a bed in the emergency department or a bed on an inpatient floor, staff should be able to quickly identify available beds. This will help reduce patient wait times and improve the overall efficiency of the hospital.

This system will also support scheduling for patient procedures and treatments, whether they take place at the bedside or in areas such as the operating room. Administrative staff must be able to schedule patients within the same system that displays bed availability. Having these features integrated within one system will make it easier for staff to manage patient movement throughout the hospital and reduce confusion when coordinating procedures.

One of the most important strategies for this project is working closely with hospital staff to understand their current workflows. By learning how staff currently track patients and manage bed assignments, the project team will be able to design the new system in a way that supports those workflows. This approach will help ensure that the system improves efficiency without disrupting the staff's normal day-to-day responsibilities.

Another important strategy will be thoroughly testing the system before it is introduced into the live environment. The project team must ensure that all system integrations, databases, and system functions are operating correctly before the system is made available to hospital staff. Following the SDLC structure and the Waterfall methodology will help reduce the likelihood of errors during implementation.

Finally, proper training and post-implementation support will be provided to staff members who will be using the system. Training will help ensure that staff feel comfortable using the new system and understand how it supports their daily tasks. In addition, post-go-live support will be available to address any issues that arise and to help staff adjust to the new system.

Risk Assessment and Mitigation Plans

When introducing a new system like this into a healthcare environment, there are always risks that may occur during the project. Any disruption to a clinician's daily tasks can have an immediate effect on patient care. Because of this, it is important that we take these risks into account early in the project and evaluate any potential issues that may arise. Our team will also create mitigation plans to address these risks and reduce the chances of them impacting the success of the project.

One issue that can occur in a project of this size is the possibility of data migration problems. Because the hospital is currently using an older and more manual system, there is a chance that some records or data may be overlooked or missing during the transition to the new system. This could create confusion for staff members if patient placements or bed assignments are inaccurate. To reduce this risk, the project team will work closely with hospital staff to perform data validation checks and confirm that backups are completed before and after the migration process.

Another risk that is commonly faced when implementing new IT systems is resistance from staff members. Introducing a new system can be intimidating, especially for staff who may not be very comfortable with technology. Some individuals may also be used to their current workflow and may be hesitant to change how they perform their daily tasks. To help address this issue, the project team will provide proper training and ongoing support during the adoption of the new system. This will help ensure that staff members feel comfortable using the system and understand how it supports their work.

The project team will also need to address potential system integration issues. The new patient tracking and scheduling system must be able to work with existing systems already in the hospital environment, such as the electronic health record (EHR), billing systems, or laboratory systems if needed. If these systems do not communicate properly, it could cause delays or incorrect information within the system. To prevent this, the team will perform integration testing before the system goes live to ensure all systems are working together properly.

Risk Assessment Matrix			
Risk	**Impact**	**Likelihood**	**Mitigation Strategy**
Data migration errors	High	Medium	Perform data backups and validation checks before and after migration
Staff resistance to new system	Medium	Medium	Provide training and support for staff members
Integration issues with other systems	Medium	Medium	Conduct integration testing before deployment

Project Timeline

Project Timeline		
Project Phase	**Duration**	**Description**
Planning	3 Weeks	Identify stakeholders and project goals
Requirements Analysis	4 Weeks	Gather system workflows and requirements
System Design	4 Weeks	Design system interface/integration and database
Development	8 Weeks	Build application and integrate systems
Testing	4 Weeks	Perform system and integration testing
Training	2 Week	Training of staff
Implementation	2 Weeks	Install system and migrate data
Support	2 Weeks	On site or Virtual Support for staff

Installation Approach

The hospital has been running on the current patient tracking system successfully for quite some time, so the best installation approach for this project will be parallel installation. With a parallel installation approach, both the old system and the new system will run at the same time. This allows the transition to be smoother and helps prevent too much disruption for the staff. One caveat to this approach is that the two systems will only run concurrently for the first week during the transition period.

While using the parallel approach, patient information will be entered into both systems so that the results can be compared by the analysts supporting the transition. This will allow the project team and stakeholders to review the results of patient locations, bed assignments, and scheduling information in real-time so that any adjustments can be made if necessary.

One of the biggest benefits of this approach is that it allows staff to gradually become familiar with the new system. At first it may seem like double the work since information will be entered into both systems, but overall the staff will benefit from seeing how the new system compares to the old one. This helps reduce stress for the staff and makes the transition to the new system much smoother.

Once the stakeholders confirm that the new patient tracking and scheduling system is functioning properly, the hospital will be ready to fully transition to the new system and discontinue the use of the older one. Using a parallel installation approach provides a safe and controlled way of introducing a new information system into a live healthcare environment.

Installation Schedule

For this transition to be completed with minimal disruptions, we will need to follow a structured installation schedule. Following a clear schedule will allow the project team to coordinate activities such as data migration, system testing, and staff training. The main purpose of this schedule is to ensure that the new system is introduced into the live hospital environment in a controlled manner.

The first step will be confirming that the hospital's infrastructure can support the new system. For example, we will need to verify that the physical hardware used by staff meets the requirements of the new system. The project team will also need to confirm network stability, server storage and availability, and ensure that all required software components are ready for the new system.

Next, the project team will migrate the existing patient data and scheduling information into the new system. During this phase, the team will also perform validation checks to confirm that patient locations and bed assignments are transferred correctly. Ensuring that this information is accurate is critical so that staff can trust the system when it goes live.

Once system testing has been completed and the data has been verified, hospital staff will receive training on how to use the new system. Training will help ensure that staff feel comfortable using the system and understand how it fits into their daily workflows.

Finally, the new system will enter the parallel installation phase, where both the old system and the new system will operate at the same time. During this period, the project team will monitor the system closely and confirm that it is functioning as expected before the hospital fully transitions to the new system.

Data Migration Plan (ETL Process)

One important part of implementing this new patient tracking and scheduling system is migrating the data from the old system into the new one. Data migration needs to be handled carefully because missing or inaccurate patient data could drastically affect the clinical workflow. Because of this, specific team members will be designated to oversee the migration process and ensure it is completed correctly. The data migration process for this project will follow the ETL method, which stands for Extract, Transform, and Load.

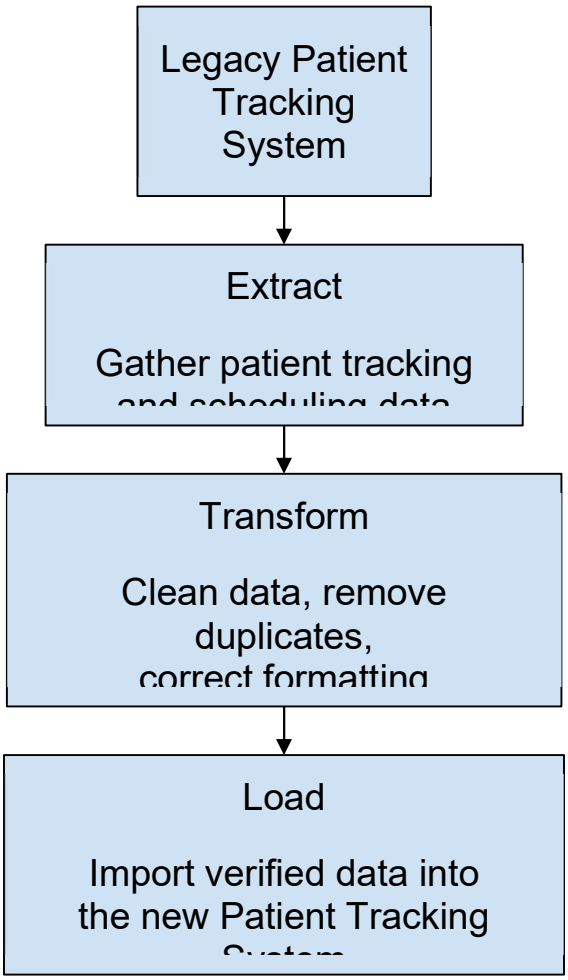
The first step is to Extract the data. During this phase, the data from the current system will be gathered by the project team and subject matter experts (SMEs). Because the hospital is currently using an outdated system, the data may exist in several different formats. For example, some of the information may be stored in Excel spreadsheets or other internal tracking tools, and in some cases it may even exist in paper documentation. The project team will need to collect all relevant patient tracking and scheduling information before moving to the next step.

The next step is to Transform the data. During this phase, the data will be reviewed and cleaned before it is moved into the new system. This may include identifying duplicate records, correcting formatting issues, and verifying that patient information is accurate. Ideally, SMEs will assist in this step because they understand the hospital workflows and can help confirm that the information is correct. Since staff will rely on this system frequently in their daily work, it is important that the data being transferred is accurate and reliable.

The final step of the ETL process is Load. During this phase, the cleaned and verified data will be imported into the new patient tracking and scheduling system. After the data has been loaded, the project team will perform validation checks to confirm that everything transferred correctly. This includes verifying patient locations, bed assignments, and scheduling information within the new system.

By following the ETL process, the project team can ensure that the data migration is completed accurately and that hospital staff will be able to rely on the information within the new system. SMEs will also assist with verifying that the migrated data appears correctly once it is available in the new environment. Proper data migration is important for ensuring a smooth transition to the new patient tracking and scheduling system.

ETL Diagram:



Testing Plan

Before the system is ready to be deployed into the environment, we will first need to perform thorough testing. This will include both integration testing and functional testing. Integration testing focuses on making sure the different applications and systems involved are communicating properly with each other. Functional testing focuses on verifying that the system itself performs the tasks it was designed to do. In addition to these tests, we will also perform User Acceptance Testing (UAT). UAT involves having a subject matter expert run through realistic scenarios using the new system to confirm that it functions the way they expect it to.

Integration testing will be completed by the systems analysts supporting the project. They will run a series of tests based on predefined scripts to determine whether the system is properly communicating with other systems in the environment. For example, the analysts may test whether patient demographic information can be pulled correctly or whether previous appointments and patient records are displayed accurately within the system.

Functional testing focuses on the system as its own application. During this stage, the project team will perform end-to-end testing to ensure that the system is operating as expected. This includes verifying that patient information can be entered correctly, beds can be assigned properly, and scheduling functions work as intended. The goal of this testing is to confirm that the system supports the daily workflow of the staff members who will be using it.

The final stage of testing is UAT. During this phase, a workflow expert will be selected to run through common tasks in a controlled environment. This allows the project team to observe how the system performs during real-world scenarios. The goal is to confirm that the new system supports the staff's workflow and does not introduce any major disruptions. Once

UAT is completed and any issues are addressed, the system will be ready to move forward with deployment.

Debugging and Error Handling Strategies

We can be as thorough as we can be in our testing, however even if all scripts and scenarios pass, we can still run into errors during the go live. In these scenarios, we need to be ready for any issues that may arise, so the project team will need to monitor the system closely and prepare to address issues if they come up quickly.

Debugging will require the analysts and developers to review any and all issues during testing and after the weeks of go-live. Keeping an eye on tickets that may come in, checking system logs and trying to recreate issues so the cause may be identified. Once the issue is found, the team will be able to quickly make a resolution and get the issue resolved.

Also, if a user enters incorrect information into our system, the system will need to provide a message to the end user that something went wrong with their input. Error handling is very important in these scenarios as well, because it alerts the user early rather than dealing with an issue later down the road.

Automated Testing Tools

Automated testing tools will also be used when testing this system. These tools are useful because they allow the project team to run multiple tests quickly without requiring a large amount of manual input. Automated testing can help verify that system updates and bug fixes do not introduce new issues into the patient tracking and scheduling system.

One example of an automated testing tool that could be used is Selenium. Selenium can simulate a user's interaction with the application, such as entering patient demographics or navigating through different screens within the system. By automating these types of actions, the project team can repeatedly test important functions of the system in a shorter amount of time.

While automated testing tools do not completely replace manual testing, they can significantly reduce the time required for repetitive testing tasks. Using automated tools alongside manual testing will help ensure that the system remains stable and reliable before it is fully deployed in the hospital environment.

Data Quality Issues

Unfortunately, we could run into data issues when migrating to the new system. There's absolutely a possibility of missing data, errors in the data, duplicates or inconsistent formatting. If these potential issues are not addressed prior to migration, it could create a lot of confusion for the staff and possibly be interrupting workflow.

To prevent this, the project team will need to perform data cleansing during the Transformation stage of the ETL process. This is where we will need to identify duplicate records, correct formatting issues or correct incorrect data that was previously entered. Cleaning and validating the data before it is loaded into the new system will help ensure that staff are working with accurate and reliable information once the system is fully implemented.

Conversion Method

Like mentioned before, for this system implementation we are going to choose the parallel conversion. Using this method, the old system and the new system will run at the same time allowing for an easier transition for the staff. Once the new system is confirmed to be

functioning correctly, and the staff is comfortable using it following training, we can stop using the old system. In a healthcare environment, this is the ideal method of using when introducing a new system.

Testing and Validation Steps

After the data has been migrated, we will need the help of the SMEs to validate that the data was transferred without error. This step is important because inaccurate data could affect patient placement and disrupt workflow. The first step in validation will be to compare the records from the old system to the data that was entered into the new system. The project team, along with the help of the SMEs, will review the data and make corrections if needed. It is best to have someone who understands the hospital's workflows review the data to ensure that the information appears accurate and makes sense within the system.

Summary of Implementation

The successful implementation of the patient tracking and scheduling system depends on several components being successful throughout the project. Each step from planning, development, testing, training and implementation plays a crucial role in the overall success and how smoothly it is introduced to the existing work environment of the hospital. Following the SDLC and Waterfall methodology keeps these goals on the right track because one step has to be completed before starting the next one.

The project team took several steps to make sure that zero risks were taken and confirm the system functions once deployed. The data migration using the ETL process and also the automatic testing assisted in these steps. The overall goal was to not disrupt the staff too much given how important their roles are in patient care.

Key Success Factors & Post-Implementation Review Criteria

The main factor to think about when judging the success of this project, is if the stakeholders are satisfied with the new system. But we also need to judge whether the system is working as expected, including tracking patient locations, monitoring open beds and the scheduling functionality is working as expected. Also, confirming that the staff who uses this software daily, are confident in their skills with using it and notice improvement in their daily workflows. After the system has been deployed, the project team will do a debrief and analyze how the system is performing. Also, evaluate how the project went and see where we could have made improvements.

Lessons Learned

The initial planning of a project this size is very important and that is a lesson I learned. Without appropriate planning, there will be poor execution of the project. It's also important to allow yourself plenty of time for training, developing, planning and post go-live support. Proper communication as the project goes on is important as well between the teams working on the project and the stakeholders. Keeping everyone informed on the progress can heavily influenc the outcome of a project this size.

Functional Requirements

The system must allow users to add patient information, view a list of patients, schedule appointments, and view scheduled appointments. The system must validate user input and display error messages when required fields are missing. The system must store patient and appointment data and allow users to retrieve that data when needed.

System Architecture and Design

For this project, I created a simple web-based application using HTML, CSS, and JavaScript. The application runs directly in the user's browser and does not require a backend server. I chose this approach to keep the system simple while still demonstrating the main functionality.

The system follows a client-side structure, where all processing and data handling happens in the browser. Patient and appointment data are stored using browser local storage, which allows the data to remain available even after refreshing the page.

The system includes basic components for user input, validation, and displaying patient and appointment data. Overall, this setup allows the system to simulate a real-world patient tracking and scheduling workflow in a simplified way.

Collaboration and Code Integration

This project was completed individually. Version control was managed using GitHub to organize project files and maintain a structured repository. All code and documentation were stored and updated within the repository to ensure consistency across the project components.

Algorithms and Data Model Implementation

The system stores patient and appointment data using simple data structures in JavaScript. Patients and appointments are stored as objects inside arrays. Each patient record includes a name, age, and reason for visit. Appointment records include the patient's name along with the date and time. The system uses straightforward logic to add patients, validate user input, store data in local storage, and display patient and appointment information.

User Interface Design and Implementation

The user interface was designed to be simple and easy to follow. The system includes sections for adding patients, scheduling appointments, and viewing data. Users enter patient information through input fields and select a patient from a dropdown when scheduling a patient appointment. The goal of the design was to keep things straightforward and reduce the number of steps needed to complete tasks. This helps simulate a more efficient workflow for healthcare staff.

Design of Tests

The system was tested using unit testing, integration testing, and basic user acceptance scenarios. Unit testing focused on individual features like adding patients and scheduling appointments. Integration testing made sure different parts of the system worked together, such as confirming that added patients appear when scheduling. User acceptance testing involved running through common workflows, like adding multiple patients and scheduling appointments, to make sure everything worked as expected.

Performance Optimization Plan

Since this is a simple prototype, performance is not a major concern. The system is designed to handle small amounts of data efficiently using local storage in the browser. If this system were expanded, improvements could include using a database, optimizing queries, and improving performance for larger datasets.

Maintenance and Support Strategy

After deployment, the system would need to be monitored and maintained to make sure it continues to work properly. This would include fixing bugs, making updates, and responding to

user feedback. Future improvements could include integrating with hospital systems, adding security features, and expanding functionality. Training and user support would also be important to ensure staff can use the system effectively.